Graph-transported factor-base arm

Fixed implementation control, 2026-10-22 PDT — Conductor T-4748, fence epoch 1. Implementation and raw controls: commit `366dd01f19`.

This report records prepared-input checks. It makes no relation-yield, rank, or ECDLP runtime inference. Source and raw values: `report.md`, `control-metrics.json`, archived P29/P37 checkpoint chains.

Exact tested objects

P29: $\frac{E}{F_{320092037}} : y^2 = x^3 + 105931039x + 284015384$, prime order 320091991, source $G = (0, 50656410)$, check vertex 8. P37: $\frac{E}{F_{81640512137}} : y^2 = x^3 + 46677163888x + 11713815639$, prime order 81640511861, source $G = (39752613976, 79465703431)$, check vertex 4. Both prefixes use archived degree-7 separable maps over their respective prime fields.

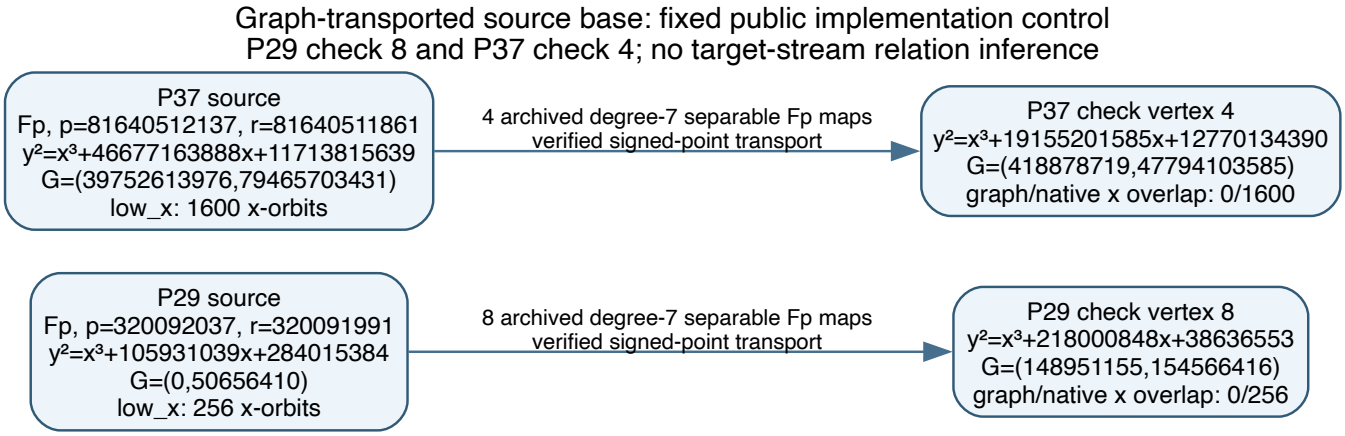

Figure 1: Archived graph transport and native-base comparisons. The two zero overlaps are fixed-input observations.

Fixed controls

The source graph base equals source native `low_x`. At P29 vertex 8, the 256-orbit graph and native bases share 0 x-orbits. At P37 vertex 4, the 1,600-orbit bases share 0. The same generator, Q, and eight public target coefficients were used in each pair. The 26 small arm/vertex preparation cases passed, including rejection of a noncheck vertex and a modified checkpoint digest. Exact raw values and checkpoint hashes are in `control-metrics.json`.

Next obligation

Bind a runnable post-freeze collector and independent count/rank verifier, then freeze the check-arm quotas and target roots under Coordinator approval. This control does not enter the prospective discovery inference.